

	STUDENT NAME			COURSE NAME	C - C, C++ & Core Java
	REGISTER. NO			TOTAL HOURS	100 HRS
	MOBILE.NO			JOINING DATE	
	CLASS TIMING			STAFF NAME	

SESSION	DATE & HOURS	MODULES	HOURS	STUDENT SIGN	STAFF SIGN
01 SESSION THEORY	DATE : _____ IN TIME : _____ OUT TIME : _____	C PROGRAM Introduction & Basics Overview of C History, installation setup keywords Datatypes Variables Identifiers	60 Mins		
02 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Input/Output & Basics Format specifiers ASCII values Constants Input/Output statements Basic practice	60 Mins		
03 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Operators - Arithmetic Operators - Assignment Operators - Relational Operators - Logical Operators - Bitwise Operators	90 Mins		
04 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Control Flow Statement - Conditional - Looping - For Loop - While - Do While - Jump statements	120 Mins		
05 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Session 1. Write a C program to find the largest among three numbers using conditional statements. 2. Write a C program to create a simple calculator using operators and control flow.	60 Mins		

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06 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Arrays • 1D, 2D arrays, operations	90 Mins		
07 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Strings • String handling, inbuilt functions	90 Mins		
08 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Functions • Types, parameters, return values	90 Mins		
09 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program 1. Write a C program to find the sum of elements in an array using a function. 2. Write a C program to reverse a string using a function. 3. Write a C program to count vowels and consonants in a string using a function.	60 Mins		
10 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Math Functions • Built-in math functions • Ceil • Floor • Sqrt	60 Mins		

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11 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Pointers • Pointer • Dynamic Memory Allocation	90 Mins		
12 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Structures & Unions • Structure • typedef • nested • Union	90 Mins		
13 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Advanced Concepts • Recursion • Storage classes • Dynamic memory	90 Mins		
14 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Derived Concepts • Combined usage (arrays, pointers, strings, structures)	90 Mins		
15 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	File Handling • File read/write, real-time usage	90 Mins		

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16 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Mini Project 1.Student Record System Add student View student Search student Delete student Store data in file	90 Mins		
17 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	C++ Introduction & Basics • Overview of C++ History Installation Keywords Datatypes Variables Identifiers	60 Mins		
18 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program • Write a program to display student details using cout. Practice variable declaration and scope.	45 Mins		
19 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Input & Output • Cin • Cout • Basic Input/Output handling • Practice programs	45 Mins		
20 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program • Read two numbers using cin and display their sum. • Create a simple calculator with cin and cout.	45 Mins		

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21 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Operators • Arithmetic • Relational • Logical • Bitwise • Assignment operators with examples	60 Mins		
22 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program • Demonstrate operator precedence with arithmetic expressions. • Implement relational checks (greater, smaller, equal).	45 Mins		
23 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Control Flow • Conditional (if, switch) • Loops (for, while) • Jump statements with practice	90 Mins		
24 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program • Write a program to check if a number is prime using loops. • Create a menu-driven program using switch.	45 Mins		
25 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Arrays • 1D Arrays • 2D Arrays • Operations and Problem-solving	60 Mins		

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26 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program • Store 10 integers in an array and find the largest. • Implement matrix multiplication.	45 Mins		
27 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Strings • String handling • Inbuilt functions • String operations	90 Mins		
28 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program • Concatenate two strings using inbuilt functions. • Count words in a sentence.	45 Mins		
29 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Functions • Function types • function overloading • default arguments • Modular coding	120 Mins		
30 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Math Functions • Built-in math functions • Logical problem solving	60 Mins		

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31 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program - Demonstrate function overloading with add(int, int) and add(double, double). - Use default arguments in a function.	45 Mins		
32 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Pointers - Pointer basics - Memory handling - Pointer arithmetic	90 Mins		
33 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program - Demonstrate pointer arithmetic with arrays. - Use new and delete for dynamic memory allocation.	45 Mins		
34 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Structures & Enum - Structure - Enumeration concepts and usage	45 Mins		
35 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program - Create a structure for employee details and display them. - Use enum for days of the week.	45 Mins		

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36 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS Basics • Class • Objects • Constructor • Destructor concepts	90 Mins		
37 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program • Create a class Student with constructor and destructor. • Demonstrate object creation and destruction.	45 Mins		
38 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Advanced OOPS • this pointer • static keyword • structs, enum • friend function	60 Mins		
39 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program • Implement a friend function to access private data. • Use static members in a class.	45 Mins		
40 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS Concepts • Inheritance • Encapsulation • Polymorphism	60 Mins		

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41 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Polymorphism Deep Dive - Function overloading - Overriding - Virtual functions	60 Mins		
42 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program - Demonstrate single and multiple inheritance. - Implement virtual functions for runtime polymorphism.	45 Mins		
43 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Advanced Concepts - Data abstraction - Namespaces - Exception handling	90 Mins		
44 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Templates Function templates and class templates	45 Mins		
45 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program - Write a function template for swapping two values. - Create a class template for a stack.	45 Mins		

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46 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	CORE JAVA Java setup Environment and Basics • Introduction To Java • Install Java and Environment Setup • Java class and Objects • JDK,JVM,JRE Scanner Class & Data Input Stream • Java Variables • Formatting Output Java Data Types Type Casting	180 Mins		
47 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program • 1)Find Maximum And Minimum Of All Datatype. • 2)Find Size And Types Of Datatype	90 Mins		
48 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Java Operators • Arithmetic operators • Assignment • Logical Relational • Bitwise Unary operators • Ternary operators & Shift Operator	180 Mins		
49 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program • 1)Add Two Numbers Without Using Arithmetic Operators. • 2).Find Largest Of Three Numbers Using Ternary Opertors. • 3).Swap Two Numbers Using Assignment. 4).Area Of Rectangle. • 5).Grade Of Student.	120 Mins		
50 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Java Control Statements • if • else if • else if • Nested if & Switch Case	180 Mins		

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51 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program 1.Calculator Using Switch. 2.Cgpa. 3.Bank Accounts. 4.Check If The Letter Is Alphabet Or Not.	120 Mins		
52 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Java Loops - While loop & Do While - For loop - Nested for ,Nested while - Break, Continue, Return	180 Mins		
53 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program 1).Pyramid Patterns. 2).Matrix Patterns. 3).Angle Patterns. 4).Zigzag Patterns. 5).Logic-Based Patterns.	120 Mins		
54 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Function - Explain about Functions - Creating a Function,Calling a Function - Types Of Arguments . - Default Parameter Value Passing a List as an Argument - Recurssion Function	180 Mins		
55 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program 1. Palindrome Check. 2.Perfect Number. 3.Count How Many Zeros In A Number. 4.Duck Number. 5.Armstrong Number.	90 Mins		

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56 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Array - What is Array. - Creating an Array&Accessing Array Elements,Length - Types of Arrays: Single-Dimensional Array, MultiDimensional Array (2D Array),Jagged Array Looping Through Arrays Passing Array to a Method Returning Array from a Method	180 Mins		
57 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program - Find First And Second Largest Number In Array. - Duplicates Array. - Merge Two Array. 3*3 Matrix. - Find The Second Largest Element.	120 Mins		
58 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	String - Define String & Acces & modify. - String vs StringBuffer vs StringBuilder - String Methods - String Comparison	180 Mins		
59 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program - Count The Vowels And Consonants In A String. - Remove Duplicates From A String. - Anagrams. - Remove Digits From A String. - First Non Repeating Character.	90 Mins		
60 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Assessment - Coding - Palindrome Check - Strongest Number	60 Mins		

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61 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:Introduction • Introduction of Oops • Class and Object • Access Modifiers • Local Scope and Global Scope	180 Mins		
62 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:Inheritance • What is Inheritance. • Types Of Inheritance • Single Inheritance • Multilevel Inheritance • Hierarchical Inheritance • Hybrid Inheritance	180 Mins		
63 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program 1. Calculate Electricity Bill. 2. Calculate Cgpa.	60 Mins		
64 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:Polymorphism • What is Polymorphism? • Types of Polymorphism • Compile-time Polymorphism • Runtime Polymorphism • Method Overloading • Method Overriding	180 Mins		
65 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program • 1. Write A Java Program To Create A Base Class Bankaccount With Methods Deposit() And Withdraw(). Create Two Subclasses Savingsaccount And Checkingaccount. Override The Withdraw() Method In Each Subclass To Impose Different Withdrawal Limits And Fees.	120 Mins		

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66 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:Abstraction & Interface - Aggregation and Interface Topics - What is Abstraction? - Abstract class Interface -Implements - Mutiple Inheritance - Difference: Abstract Class vs Interface	180 Mins		
67 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program - Prime Number. - Linear Search.	60 Mins		
68 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:Encapsulation - What is Encapsulation? - Getter & Setter - This keyword - super Keyword - Final Keyword	180 Mins		
69 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program Write A Java Program To Create A Class Called Circle With A Private Instance Variable Radius. Provide Public Getter And Setter Methods To Access And Modify The Radius Variable. However, Provide Two Methods Called Calculatearea() And Calculateperimeter() That Return The Calculated Area And Perimeter Based On The Current Radius Value.	90 Mins		
70 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:Constructor - What is Constructor? - Default Constructor - Parameterized Constructor - constructor Overloading - Copy Constructor	180 Mins		

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71 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program • Find The Number Of Days In A Month.	60 Mins		
72 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:Exception handling • What is Exception Handling? • Keywords in Exception Handling Types of Exceptions Multiple Catch Blocks Nested try Block throw vs throws Custom Exceptions	120 Mins		
73 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program 1. Write A Java Program To Create A Method That Takes An Integer As A Parameter And Throws An Exception If The Number Is Odd. 2. Write A Java Program That Reads A List Of Integers From The User And Throws An Exception If Any Numbers Are Duplicates.	120 Mins		
74 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:Wrapper Classes & Type Safety • Primitive to Object Conversion • Autoboxing & Unboxing • Using Integer, Double, Boolean, Character	180 Mins		
75 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program 1. Write A Java Program To Create A Method That Takes An Integer As A Parameter And Throws An Exception If The Number Is Odd. 2. Write A Java Program That Reads A List Of Integers From The User And Throws An Exception If Any Numbers Are Duplicates.	120 Mins		

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76 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:MultiThreading - Creating Threads (Thread Class & Runnable Interface) - Thread Lifecycle - Synchronization (synchronized Keyword) Inter-Thread Communication (wait(), notify()) - Executor Framework	180 Mins		
77 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:Collection - List Interface (ArrayList, LinkedList, Vector) - Set Interface (HashSet, LinkedHashSet, TreeSet) - Map Interface (HashMap, LinkedHashMap, TreeMap) - Queue Interface (PriorityQueue, Deque)	180 Mins		
78 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program - Write A Java Program To Update An Array Element By The Given Element. - Write A Java Program To Create A Reverse Order View Of The Elements Contained In A Given TreeSet. - Write A Java Program To Associate The Specified Value With The Specified Key In A Hashmap.	120 Mins		
79 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:Sorting & Comparable/Comparator - Sorting Lists and Arrays (sort(), reverseOrder()) - Custom Sorting with Comparable and Comparator	120 Mins		
80 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program Adding, removing, and retrieving elements	90 Mins		

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81 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:File Handling in Java (I/O) - File Reading and Writing (FileReader, FileWriter) - BufferedReader & BufferedWriter - Serialization & Deserialization	60 Mins		
82 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Practice Program Create a File, Write Data, and Read it Back	60 Mins		
83 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	OOPS:Lambda Expressions (Java 8) - Understanding Functional Programming () -> {} Syntax - Using Lambda for Collections	60 Mins		
84 SESSION PRACTICAL	DATE : _____ IN TIME : _____ OUT TIME : _____	Assessment ATM Machine	60 Mins		